

Ari M. Friedman

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EDUCATION

University of Florida Honors, Herbert Wertheim College of Engineering
B.S. in Biomedical Engineering, Honors

Gainesville FL
Expected May 2028

Minor in Business Administration

- GPA: 3.94/4.0

Relevant Coursework: Biomedical Instrumentation, Biosignals & Systems, Cell & Systems Physiology, Computer Applications for BME, Differential Equations, Engineering Mechanics: Statics, Physics (E&M)

WORK EXPERIENCE

Unmanned Aircraft Systems Laboratory

May 2026 – Present

Research Assistant

- Designed a custom Printed Circuit Board (PCB) and Power Distribution Board (PDB) to distribute motor power and flight-control signals for autonomous wildfire detection and suppression drone systems

Human Neuromechanics Laboratory

May 2025 – Present

Research Assistant

- Processed EMG, 3D motion capture, and force plate data in Motive, Visual3D, and MATLAB to assess lower-limb biomechanics during post-ankle replacement gait recovery
- Collected data for human-subject experiments using wearable EEG and IMU systems, including participant preparation, sensor placement, and structured experimental data acquisition
- Created CAD designs that enhanced accuracy of beam size adjustments and placements in the experimental setup of a visual perturbation balance training study
- Engineered an HTML-based application to log and export time-stamped beam-walking data to Excel, enabling efficient step-off frequency analysis across 200+ experimental data collections

LEADERSHIP & INVOLVEMENT

Dream Team Engineering

August 2024 – Present

Executive Board Member – Director of Recruitment & Design Team Member

- Developed individualized shadowing cycle plans for 26 vetted engineering students, aligning placements with technical interests and project needs across multiple multidisciplinary teams
- Orchestrated weekly meetings, technical workshops with guest speakers, and social events to strengthen members' skill sets and foster a cohesive cohort
- Designed a functional liver transplant model using SolidWorks to develop a medical training device for resident physicians at UF Health Shands Hospital

Generational Relief in Prosthetics

August 2025 – April 2026

Research and Development Team Captain

- Directed an 8-person multidisciplinary R&D team, organizing technical design reviews, delegating prototyping tasks, and coordinating milestones to support iterative testing and device reliability
- Led development of an assistive neuromuscular interface, integrating Arduino-based EMG acquisition with Python algorithms to translate muscle activity into real-time electronic music control signals
- CAD-designed enclosure and soldered circuitry to reduce electrical noise and improve signal quality

INTERESTS, CERTIFICATIONS & SKILLS

- **Skills:** Arduino, C++, Fusion 360, MATLAB, KiCAD, Motive, Python, SolidWorks, Visual3D
- **Certificate:** SolidWorks CAD Fundamentals
- **Interests:** Sound Design & Music Production, Strength Training, Swimming, Playing Soccer
- **Scholarships:** University of Florida Presidential, Balanced Man, Solon Academic and Athletic Boosters